S2.2 The covalent model

Multiple covalent bonds

Learning outcomes

By the end of this section you should be able to:

- Deduce the Lewis formula for organic and inorganic molecules and ions with double and triple bonds present.
- Explain the relationship between the number of bonds and bond length and bond strength.

Covalent bonds are like an electrostatic glue that holds atoms together in a molecule.

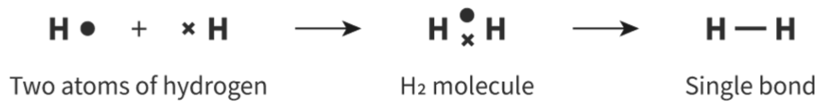
In each of the molecules you have examined in previous sections, one pair of electrons has been shared between two atoms. But sometimes, one pair of electrons is not enough and atoms may need to share two or even three pairs of electrons by forming two or even three bonds. Why do you think this is?

This section will explore how the number of pairs of electrons shared in a covalent bond determines particular properties of the bond, such as the length and strength of the bond.

Single, double and triple bonds

As we learned in section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/), each covalent bond formed between atoms contains a pair of electrons.

Hydrogen molecules, H_2, contain a single bond (**Figure 1**). This means one pair of electrons are being shared, two electrons in total. That is because hydrogen only requires one pair of electrons to be shared to fill its valence shell. Fluorine molecules, F_2, also contain a single bond, one pair of electrons is being shared between each fluorine atom. This is enough for each fluorine atom to fulfil the octet rule.



More information

The diagram illustrates the formation of an H2 molecule from two separate hydrogen atoms. On the left, there are two hydrogen atoms represented with symbols: "H • + × H," indicating single electrons. An arrow points to the middle of the image, showing the result as "H2 molecule" with symbols: "H × H," representing the shared electrons in the molecule. Another arrow leads to the right-most part, labeled "Single bond," depicting the bond as "H—H," illustrating the single covalent bond between the two hydrogen atoms.

[Generated by AI]

Figure 1. Single bonds form in hydrogen and fluorine molecules.

More information for figure 1

The diagram shows the formation of a single covalent bond between two fluorine atoms. On the left, two separate fluorine atoms are displayed, each represented with 'F' symbols and surrounded by electron dot structures. Each atom has seven electrons in the outer shell. They move towards each other, as indicated by an arrow, to share one pair of electrons, forming a single bond. The middle section shows the resulting F2 molecule where the electron dots overlap, indicating electron sharing. The right side presents the simplification of the bonding into a single dash connecting the two 'F' symbols, representing the single covalent bond formed. Labels beneath each section read: 'Two atoms of Fluorine,' 'F2 molecule,' and 'Single bond,' respectively.

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What happens if the sharing of one pair of electrons is not enough to fulfil the octet rule?

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To understand this, look at the oxygen molecule, O_2. Think about the number of electrons in the valence shell of one oxygen atom. If two oxygen atoms share one pair of electrons and form a single bond, will that fulfil the octet rule?

Count the number of valence electrons in the Lewis formula in **Figure 2** to deduce an answer to this.

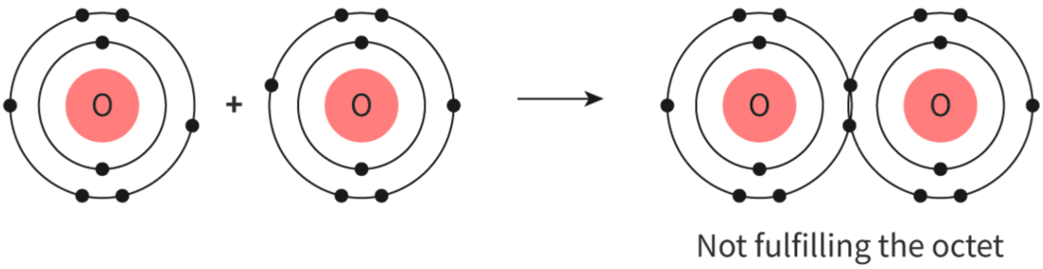

Figure 2. Oxygen cannot fulfil the octet rule and reach stability by sharing just one pair of electrons.

More information for figure 2

The diagram illustrates two oxygen atoms each depicted with their electron shells. The first and second images show separate oxygen atoms with electrons arranged in shells around the nucleus. The third image shows two oxygen atoms connected by a single bond, each sharing one pair of electrons. Despite this sharing, the note 'Not fulfilling the octet' indicates that neither atom achieves a full octet of electrons. This exemplifies why a single bond is insufficient for oxygen to reach stability, highlighting the necessity for a double bond to share two pairs of electrons and fulfill the octet rule.

[Generated by AI]

Oxygen molecules cannot form with just a single bond as it would not fulfil the octet rule with the sharing of just one pair of electrons. Instead, it must form a double bond where it shares two pairs of electrons and four electrons in total – we call this a **double bond**. This can be seen in the Bohr diagram in **Figure 3** and the Lewis formula in **Figure 4**.

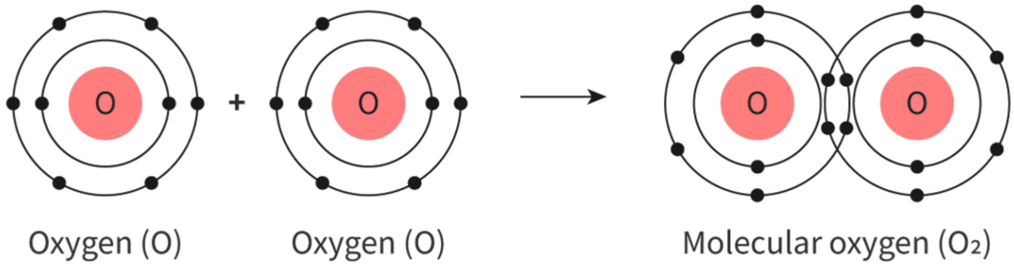

Figure 3. Bohr diagram showing the formation of an oxygen molecule containing double bonds, formed from two oxygen atoms.

More information for figure 3

The Bohr diagram illustrates how an oxygen molecule forms with double bonds. On the left and center are two separate oxygen atoms, each with a nucleus labeled "O" and surrounded by electron shells depicting the electrons. Each atom has six electrons in its outer shell, represented by dots. The atoms are shown moving towards each other, indicated by an arrow. On the right, the two oxygen atoms are combined to form a molecular oxygen (O2) with electron sharing shown in the overlapping outer shells, demonstrating the double bond formation where there are two pairs of shared electrons between the oxygen atoms.

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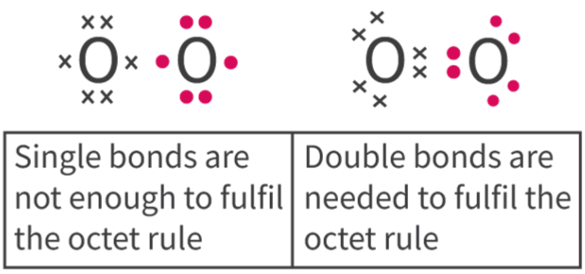

Figure 4. Lewis diagrams showing that oxygen must form a double bond to fulfil the octet rule.

More information for figure 4

The image contains two Lewis diagrams illustrating bonding in oxygen molecules. The first section depicts an oxygen molecule with single bonds, shown as two oxygen atoms connected by one line. Each oxygen atom is surrounded by dots and crosses, representing valence electrons, with a total of seven around each atom. Below this diagram, there is text stating, "Single bonds are not enough to fulfil the octet rule."

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The second section illustrates an oxygen molecule with a double bond. Here, the two oxygen atoms are connected by two lines, representing a double bond. Each oxygen atom has a total of eight electrons around it, fulfilling the octet rule. The text beneath this diagram says, "Double bonds are needed to fulfil the octet rule." These diagrams clearly denote the necessity of double bonds in achieving stable electron configurations for oxygen atoms.

[Generated by AI]

What happens when we examine a molecule of nitrogen? Like oxygen molecules, nitrogen molecules cannot fulfil the octet with the sharing of one, or even two pairs of electrons. Instead, nitrogen molecules, N_2, must form triple bonds (**Interactive 1**), sharing three pairs of electrons or six electrons in total.

More information

The image is a diagram illustrating the formation of a nitrogen molecule, N2, through triple bonding. On the left side, two separate nitrogen atoms are shown, each represented by the letter 'N' surrounded by five red dots indicating valence electrons. Between the two nitrogen atoms, plus sign and an arrow point toward the resulting nitrogen molecule. The central part of the diagram displays the two nitrogen atoms connected by three lines symbolizing triple bonds. The depiction emphasizes the sharing of six electrons in total. This is further represented alternatively by three horizontal red lines between 'N' letters.

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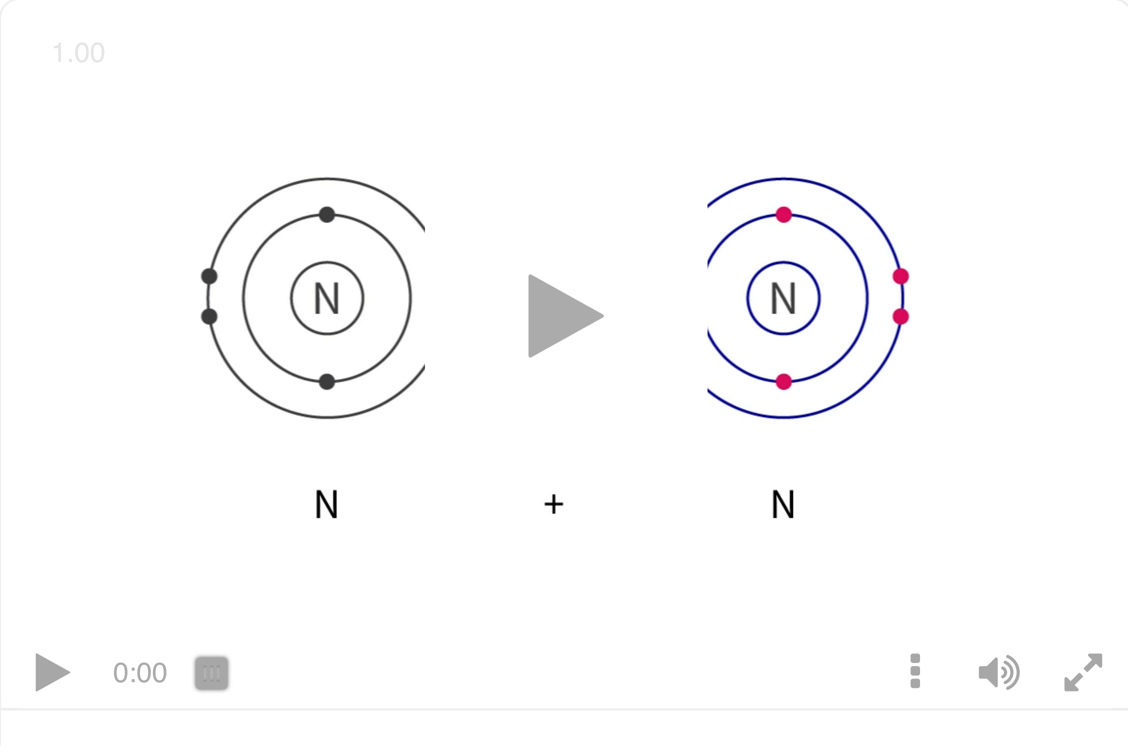


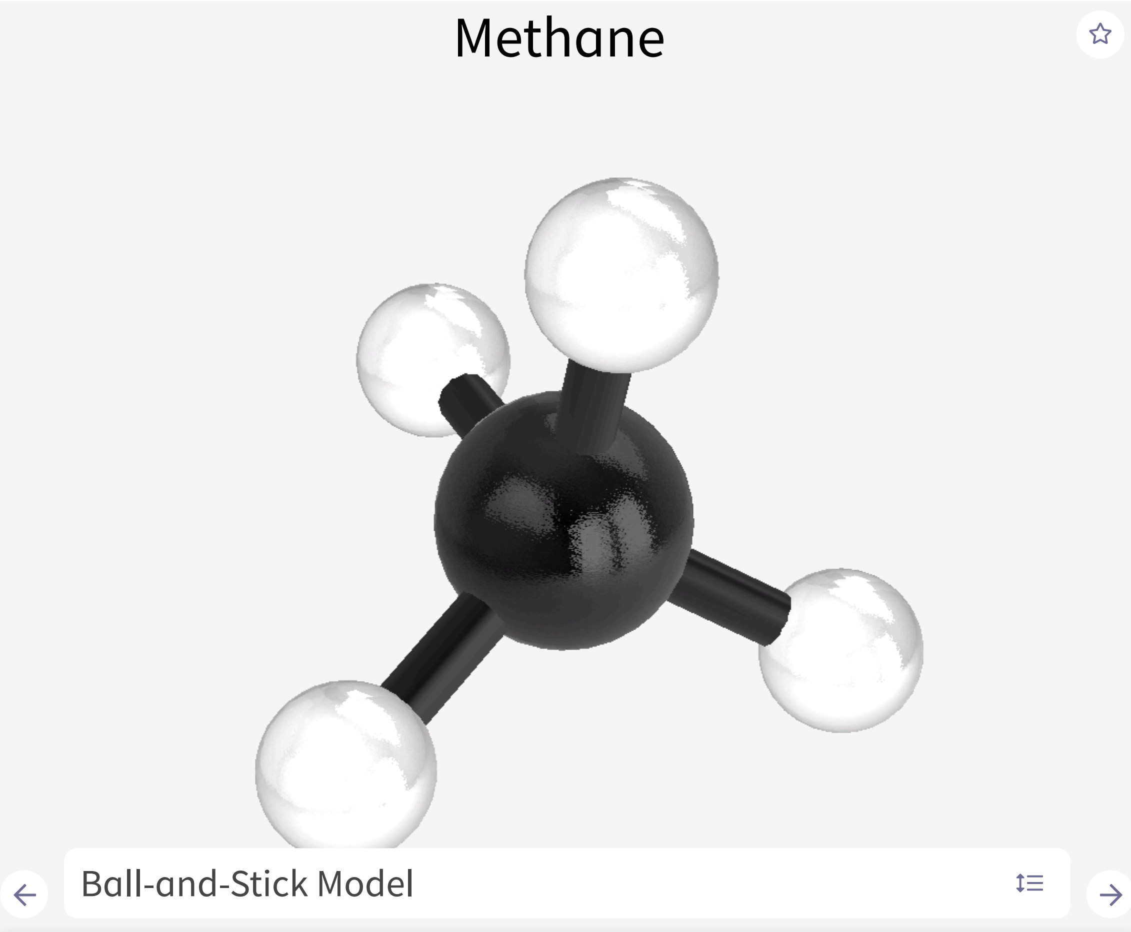
Interactive 1. Bohr and Lewis Formulas Showing the Formation of Triple Bonds in Nitrogen Molecules.

More information for interactive 1

An interactive video depicts the Bohr model for the formation of a triple bond in the nitrogen molecule. Two nitrogen atoms interact to form a nitrogen molecule (N_2).
In the Bohr model of nitrogen, each nitrogen atom has two electron shells: the first (inner) shell contains two electrons, while the second (outer) shell has five electrons.
As the two nitrogen atoms move closer, their outer shells overlap. Each atom shares three of its outer electrons, resulting in a triple covalent bond. In total, six electrons are shared in the overlapping region.
This video helps users understand how a triple bond forms in a nitrogen molecule.

Being able to visualise these molecules as three-dimensional structures is an important skill for understanding how they react in later subtopics, but this can be difficult.

Use these models in **Interactive 2** to help you do this. First explore the two-dimensional 'structural formula' of each one and then switch to the 'ball and stick' model. Practice rotating the molecule to recognise the three-dimensional reality of molecules.



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Interactive 2. Structure of Methane.

More information for interactive 2

An interactive illustration displays different representations of methane. Users can rotate and move these representations in any direction. A carousel selector with arrows is at the bottom, allowing switching between a ball-and-stick model, a sphere model, and a structural formula. The carousel also includes options to highlight carbon, hydrogen, and molecular bonds in the ball-and-stick model and options to highlight carbon and hydrogen in the sphere model.

In the ball-and-stick model, a large central sphere representing carbon is bonded to four smaller spheres representing hydrogen atoms. Hydrogen atoms are connected to the carbon atom by thin rods, representing single bonds. This structure forms a tetrahedral geometry. The sphere model depicts the same arrangement but without the connecting rods. In the structural formula, the central C atom is connected to H atoms at the top, right, bottom, and left via single bonds. This representation indicates the two-dimensional structure of methane.

On the right side, options for changing the background color and option for zooming the interactive are given.

Below the selector are options labeled introduction, share, capture, augmented reality, and language.

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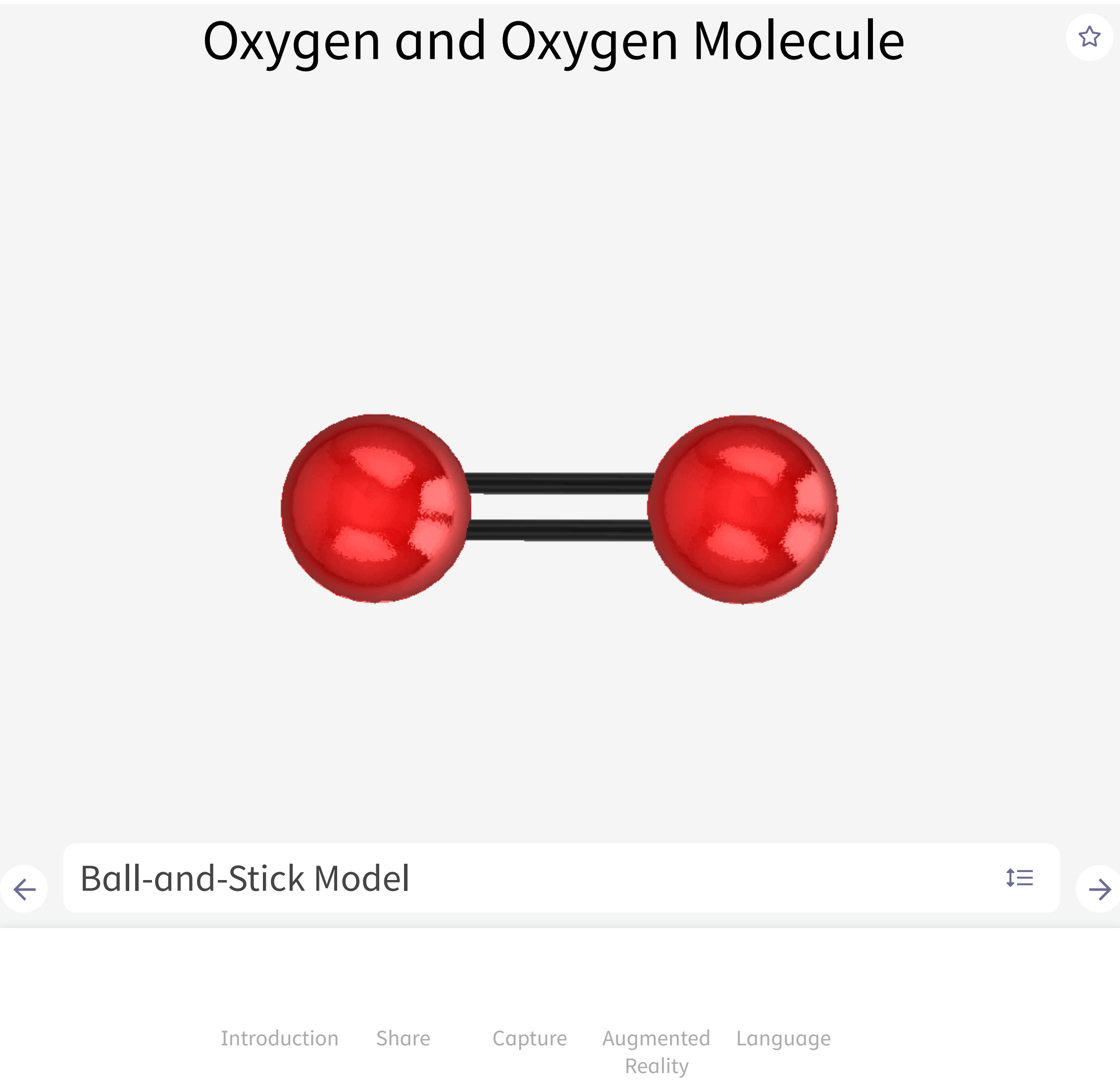
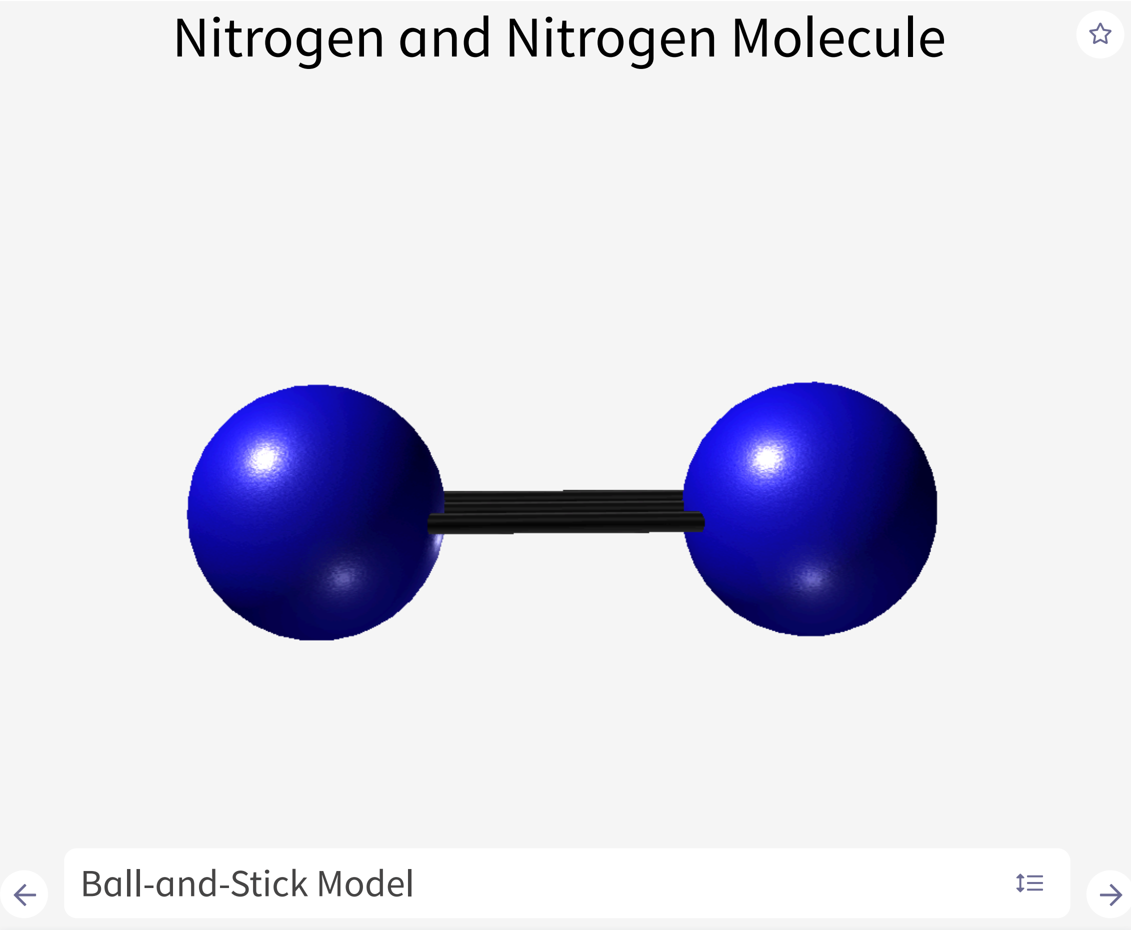
The introduction option provides the following details:
Molecular formula: CH_4
Molar mass: 16.04 g/mol
Description: Methane is the simplest alkane hydrocarbon. Under standard conditions, it is a colorless, non-toxic, and odorless gas that is lighter than air. Methane is highly combustible and can form an explosive mixture when combined with air.
Applications: Methane is commonly used as a heat source in both industrial and household applications. It is also an important chemical raw material used in the production of various compounds, such as methanol, acetylene, hydrogen, and cyanide.
Occurrence: Methane is the primary component of natural gas (approximately 99%) and is also present in coal gas (around 30%).
These three-dimensional representations allow users to explore methane, including its bonding and geometry. They provide a clear view of the spatial arrangement of atoms in the molecule.



Interactive 2. 3D Structure of Oxygen Molecule.

More information for interactive 2

An interactive illustration displays different representations of the oxygen molecule. Users can rotate and move these representations in any direction. A carousel selector with arrows is at the bottom right, allowing users to switch between a ball-and-stick model, a sphere model, and a formula. The carousel also includes options to highlight oxygen and molecular bonds in the ball-and-stick model and option to highlight oxygen in the sphere model.
In the ball-and-stick model, two spheres representing oxygen atoms are connected by two thin rods.
The sphere model shows two fused oxygen spheres without any connecting rods.
In the formula, the two O atoms are connected via a double bond.
Below the selector are options labeled introduction, share, capture, augmented reality, and language.
The introduction option provides the following details:
Symbol: O
Atomic number: 8
International name: Oxygen
General formula: O_2
Relative Atomic Weight: 15.999
Description: Oxygen is a colorless, tasteless, and odorless gas. When liquefied, it has a bluish hue. Oxygen atoms are unstable in their atomic form and naturally combine to form diatomic molecules (O_2), with a double bond between atoms. It is classified as a typical non-metal.
Applications: Oxygen is widely used in the production of iron and steel, for welding and cutting metals, in the glass industry, and in medicine (breathing apparatus). It is also used as an oxidizer in rocket fuel.
Occurrence: Oxygen is the most abundant element on Earth, present in the atmosphere, hydrosphere, and Earth's crust. It is a biogenic element, forming a fundamental part of all living organisms.
These three-dimensional representations allow users to explore the O_2 molecule, including its bonding and geometry. They provide a clear view of the spatial arrangement of atoms in the molecule.



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Interactive 2. 3D Structure of Nitrogen Molecule.

More information for interactive 2

The interactive illustration presents various representations of nitrogen and the nitrogen molecule. Users can rotate and manipulate these models freely to explore different perspectives. A carousel selector with navigation arrows at the bottom allows switching between three views:

1. Ball-and-Stick Model — Two spheres representing nitrogen atoms are connected by three thin rods, symbolizing the triple bond in N_2.
2. Sphere Model — Displays two fused nitrogen spheres without visible connecting rods, emphasizing the molecule's compact nature.
3. Formula Representation — Shows two "N" symbols linked by three horizontal lines (≡), indicating a triple covalent bond.

The carousel also provides options to **highlight nitrogen atoms and molecular bonds** in the ball-and-stick model for better visualization.
Below the selector, several options are available, such as introduction, share, capture, augmented reality, and language.
The introduction option provides the following details:

Symbol: N
Atomic number: 7
International name: Nitrogen
General formula: N_2
Relative Atomic Weight: 14.007
Description: Nitrogen is a colorless, non-toxic, inert gas that is tasteless and odorless. It exists as diatomic molecules N_2, with a triple bond between the nitrogen atoms. The gas is a typical non-metal.
Applications: Nitrogen is produced through fractional distillation of liquefied air and is often used as an inert atmosphere for various chemical reactions. It is a key raw material in the chemical industry, especially for the production of ammonia and nitric oxide. Ammonia is subsequently used to manufacture nitric acid and fertilizers. Liquid nitrogen, with a boiling point of $-196^{\circ}C$, is used as a cooling agent in the medical and food industries.
Occurrence: Elemental (free) nitrogen forms 78% of the Earth's atmosphere. In its bound form, nitrogen is found in various compounds such as Chilean nitrate ($NaNO_3$), ammonia (NH_3), and numerous organic compounds. Nitrogen is a biogenic element, essential for life, contained in proteins and nucleic acids.
These three-dimensional representations allow users to explore the N_2 molecule, including its bonding and geometry. They provide a clear view of the spatial arrangement of atoms in the molecule.

As the chemical properties of the element is largely due to the valence electrons rather than the inner electrons, you will always be asked to draw Lewis formulas. To correctly represent the formation of single, double and triple bonds on a Lewis formula, add in additional lines, dots or crosses to show these extra electrons in multiple bonds, as you saw with the oxygen and nitrogen molecules before.

To determine the number of single, double or triple bonds required in each molecule, you should follow the rules outlined previously in this section, but include the two additional last steps.

Drawing Lewis formulas for molecules and ions with multiple bonds

1. Calculate the total number of valence electrons in the molecules (this is the number of valence electrons multiplied by the

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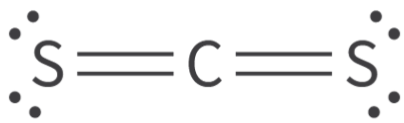
number of each atom).

2. Identify the central atom using the rules, we covered in section 2.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/). Draw the skeletal structure of the molecule by linking the peripheral atoms to the central atom with single bonds.
3. Add remaining electrons (following the octet rule) as non-bonding pairs.
4. If there are not enough electrons to complete the octets, use one or more of the non-bonding pairs to form double and triple bonds.
5. Check that the total number of electrons in the finished structure is equal to your calculation in the first step.

Worked example 1

Identify the number of bonding and non-bonding pairs of electrons in a molecule of carbon disulfide.



More information

The image shows the molecular structure of carbon disulfide (CS2). The central carbon (C) atom forms two double bonds with two sulfur (S) atoms, located on either side. Each sulfur atom has two lone pairs of electrons, depicted as two pairs of dots around the sulfur atoms. The image illustrates the linear arrangement of the atoms in the molecule, with the carbon atom in the middle linked to each sulfur with double bonds, and the non-bonding pairs clearly shown.

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There are 8 bonding electrons (Each line represents a pair of electrons) and 4 non-bonding pairs of electrons (2 pairs on each sulfur)

Study skills

Be careful when answering questions that you read and interpret them properly. Questions can ask for **either** how many electrons are shared or how many pairs of electrons are shared between two atoms.

Worked example 2

Deduce the Lewis formula for a molecule of carbon dioxide, CO_2.

Solution steps	Ca
Step 1: Calculate total valence electrons.	C
Step 2: Identify the central atom - here it will be carbon as it is further left on the periodic table than oxygen, it is the least numerous (only 1 carbon compared to 2 oxygens) and is written first in the chemical formula. Join all peripheral atoms (oxygens) to the central atom (carbon) with single bonds.	
Step 3: Add remaining electrons non-bonding pairs.	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. *(Carbon was not fulfilling the rule otherwise.)*	

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Solution steps	Ca
Step 5: Check number of electrons added is as calculated in Step 1.	C ea bo

Worked example 3

Deduce the Lewis formula for the cyanide ion, CN^{-}.

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\mathrm{C} + \mathrm{N} + 1$ (negative cl
Step 2: There are only two atoms here so no central atom to identify. Join both atoms with single bonds.	
Step 3: Add remaining electrons as non-bonding pairs.	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule (one non-bonding pair moved from each atom to make additional bonds).	
Step 5: Check number of electrons added is as calculated in Step 1.	C = 3 bonds + 1 non-bon N = 3 bonds + 1 non-bon

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Solution steps	Calculations
	

Comparing single, double and triple bonds

Single, double and triple bonds differ in how many pairs of electrons are being shared (**Figure 5**). At the beginning of this section, we discussed covalent bonds as being the ‘electrostatic glue’ that holds a molecule together. How do you think the number of bonds (single, double or triple) may affect the properties of the bond, for example the bond strength?

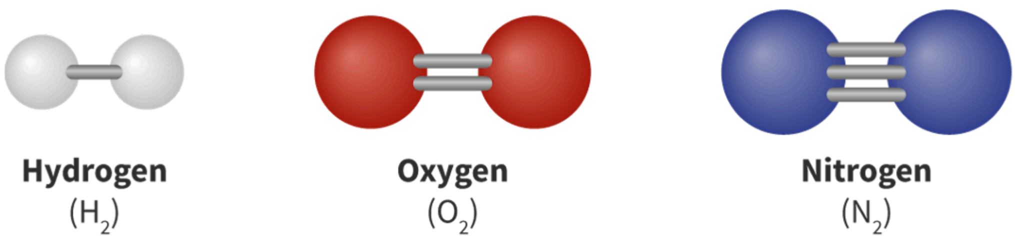


Figure 5. Single, double and triple bonds in hydrogen, oxygen and nitrogen molecules.

More information for figure 5

The image is a diagram illustrating molecular structures of hydrogen, oxygen, and nitrogen. Each structure shows different types of chemical bonds.

- **Hydrogen (H_2):** Depicted as two white spheres connected by one gray bar representing a single bond.

- **Oxygen (O_2):** Shown as two red spheres connected by two gray bars indicating a double bond.
- **Nitrogen (N_2):** Consists of two blue spheres connected by three gray bars demonstrating a triple bond.

The diagram visually distinguishes the molecules by color and bond type, highlighting how they differ in the number of shared electron pairs, ultimately affecting bond strength.

[Generated by AI]

Bond strength is a measure of the energy required to break a bond. As you may have guessed, the more bonds there are, the stronger the bond. Double and triple bonds have a greater bond strength than single bonds, with triple bonds being stronger than double bonds (**Figure 6**). Multiple bonds involve a greater number of shared electrons so the electrostatic attraction between them and the positive nuclei is stronger, hence a stronger bond.

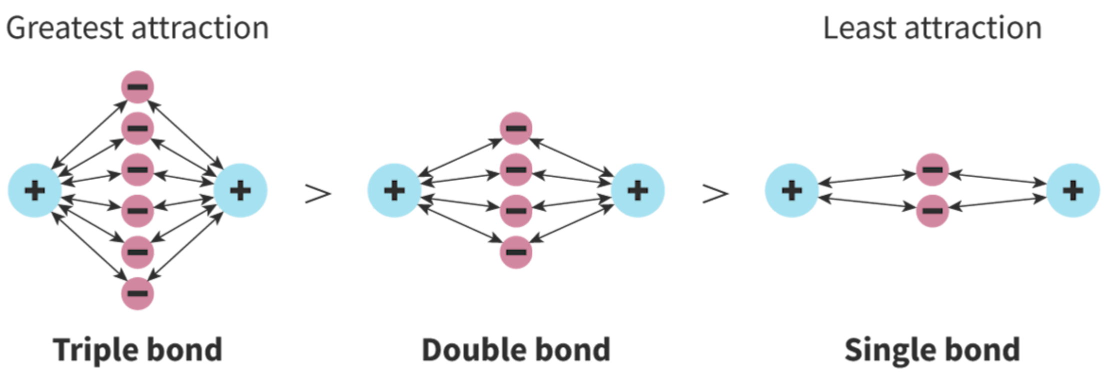


Figure 6. Triple bonds are the strongest bonds, followed by double bonds. Single bonds are the weakest.

More information for figure 6

The image is a diagram illustrating the relative strength of single, double, and triple bonds. It shows three different molecular structures labeled as "Single bond," "Double bond," and "Triple bond." Each structure consists of blue circles marked with a plus sign, representing nuclei, connected by lines to pink circles marked with a minus sign, indicating electron pairs. The number of connections increases from the single to the triple bond, demonstrating the concept of bond strength.

The triple bond has the greatest number of connections, with three pairs of pink circles between two nuclei, illustrating the greatest attraction. The double bond has two pairs, indicating a medium level of attraction, while the single bond has one pair, representing the least attraction. Arrows or lines indicate connection and attraction strength between the components, emphasizing the electrostatic attraction involved. This visual representation helps explain how multiple bonds involve a greater number of shared electrons, leading to stronger bonds due to increased electrostatic attraction between the electrons and the positive nuclei.

[Generated by AI]

The bond type will also affect another property, bond **length**. This is defined as a measure of the distance between two bonded nuclei. Using the electrostatic attraction as a 'glue' analogy again, which bonds do you think would be longest and which the shortest?

Double and triple bonds are shorter than single bonds as the same electrostatic attraction that makes the bond hard to break also pulls the nuclei closer together. As single bonds only have one pair of electrons holding them together, they are less able to pull the two nuclei together.

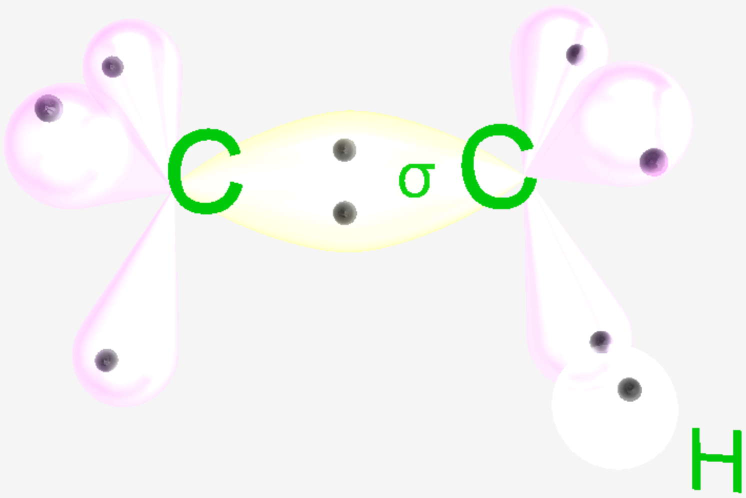
However, not all single bonds are the same length or strength. The same is true for double and triple bonds. Can you think of any factors that could affect these values?

Study skills

The values for bond length and strength can be found in the DP chemistry data booklet in section 11 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-or-average-covalent-bond-lengths-id-45113/) and section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/), respectively.

Use the models in **Interactive 3** to help you better visualise the difference between single, double and triple bonds.

Electron Density Distribution of Ethane Molecule Single Bond



Entire model

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Interactive 3. Electron Density Distribution of Ethane Molecule Single Bond.

More information for interactive 3

An interactive animation illustration exhibits the electron density distribution of the ethane molecule's single bond. It displays two sp^3 carbon atoms, each with four lobes forming a tetrahedral geometry. Each lobe has one electron. These two carbon atoms interact, with one lobe from each carbon overlapping to form a sigma bond between C—C.

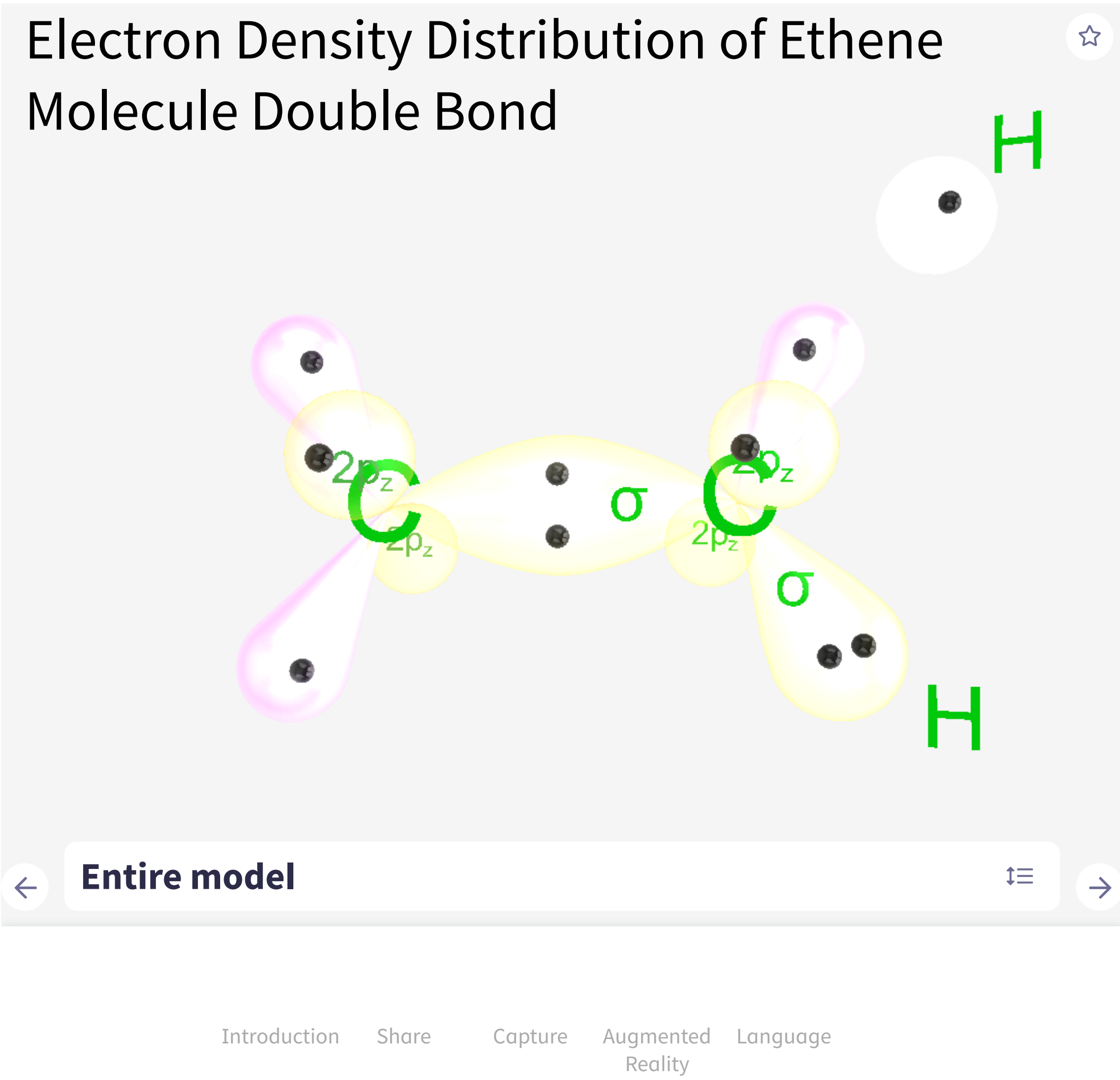
Three hydrogen atoms, each with an s orbital containing one electron, interact with the remaining three orbitals of each carbon atom, forming sigma bonds between C—H.

A carousel selector with arrows is at the bottom, displaying the entire model. Users can rotate the model to view it from different angles. Below is a seek bar with a play or pause button, allowing users to demonstrate the steps of bond formation.

At the bottom, introduction, share, capture, augmented reality, and language options are available.

The introduction option displays the following details:
The sp^3 hybridization is used to describe the bonding in the ethane molecule. When a bond is formed between carbon atoms, two sp^3 orbitals overlap to form a σ-bond. This results in a simple bond, consisting of one σ-bond.
When a bond is formed between a carbon atom and a hydrogen atom, the sp^3 orbital and the s-orbital overlap to form a σ-bond, resulting in a simple bond.

This animation helps users understand the hybridization of carbon, bonding in ethane, and electron density distribution.



Interactive 3. Electron Density Distribution of Ethene Molecule Double Bond.

More information for interactive 3

An interactive animation illustration exhibits the electron density distribution of the ethene molecule double bond. It displays two sp^2 hybridized carbon atoms in a trigonal planar geometry. Each sp^2 carbon has three sp^2 lobes in the trigonal plane and a $2p_z$ unhybridized lobe perpendicular to the plane. Each lobe contains one electron. These two carbon atoms interact, with one sp^2 lobe from each carbon overlapping to form a sigma bond between C—C.
Two hydrogen atoms, each with an s orbital containing one electron, interact with the remaining two sp^2 orbitals of each carbon atom, forming sigma bonds between C—H.
The unhybridized $2p_z$ orbitals on each carbon overlap sideways, forming a pi bond, resulting in a C=C double bond.
A carousel selector with arrows is at the bottom, displaying the entire model. Users can rotate the model to view it from different angles. Below is a seek bar with a play or pause button, allowing users to demonstrate the steps of bond formation.
At the bottom, introduction, share, capture, augmented reality, and language options are available.
The introduction option displays the following details:
The sp^2 hybridization is used to describe the bonding in the ethene (ethylene) molecule. When a bond is formed between carbon atoms, two sp^2 orbitals overlap to form a σ-bond, while the unhybridized p-orbitals of each carbon atom overlap above and below the plane of the bonded atoms to form a single π-bond. This results in a double bond, consisting of one σ-bond and one π-bond.
When a bond is formed between a carbon atom and a hydrogen atom, the sp^2 orbital and s-orbital overlap again to form a σ-bond, resulting in a simple bond.
This animation helps users understand the hybridization of carbon, bonding in ethene, and electron density distribution.

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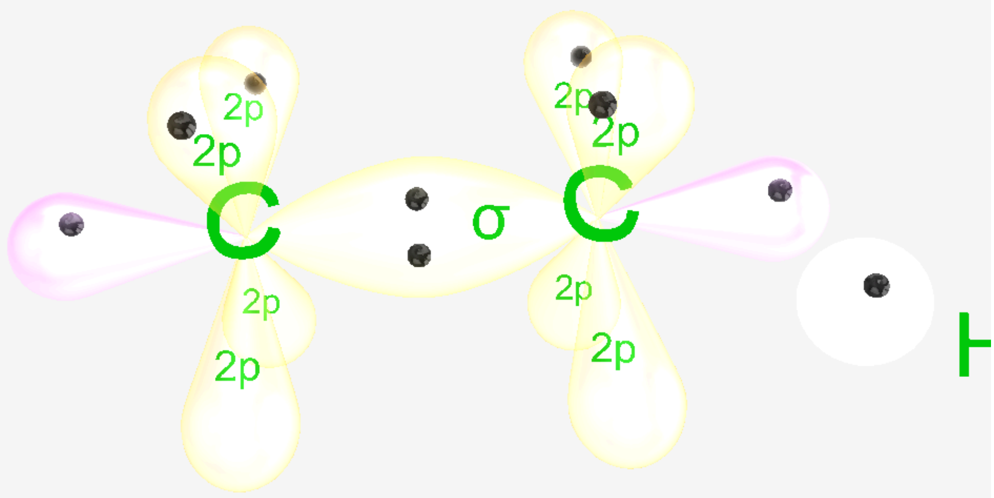
Electron Density Distribution of Ethyne Molecule Triple Bond



Entire model

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Interactive 3. Electron Density Distribution of Ethyne Molecule Triple Bond.

More information for interactive 3

An interactive animation illustration exhibits the electron density distribution of the ethyne molecule triple bond. It displays two sp-hybridized carbon atoms in a linear geometry. Each sp carbon has two sp lobes in a straight line and two unhybridized 2p lobes oriented perpendicularly to each other. Each lobe contains one electron.
These two carbon atoms interact, with one sp lobe from each carbon overlapping to form a sigma bond between C—C.
One hydrogen atom, with an s orbital containing one electron, interacts with the remaining sp orbital of each carbon atom, forming sigma bonds between C—H.
The two unhybridized 2p orbitals on each carbon overlap sideways, forming two pi bonds, resulting in a C≡C triple bond.
A carousel selector with arrows is at the bottom, displaying the entire model. Users can rotate the model to view it from different angles. Below is a seek bar with a play or pause button, allowing users to demonstrate the steps of bond formation.
At the bottom, introduction, share, capture, augmented reality, and language options are available.

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The introduction option displays the following details:
Sp hybridization is used to describe the bonding in the ethylene (acetylene) molecule.
When a bond is formed between carbon atoms, two sp orbitals overlap to form a single σ-bond, and the remaining four non-hybrid p-orbitals overlap to form two π-bonds, resulting in a triple bond consisting of one σ-bond and two π-bonds. The π-bond electron distribution creates a cylindrical electron density around the bond axis.
When a bond is formed between a carbon atom and a hydrogen atom, the sp orbital and s-orbital overlap again to form σ-bonds (simple bonds are formed).
This animation helps users understand the hybridization of carbon, bonding in ethyne, and electron density distribution.

You can now apply this knowledge in the activity in **Interactive 4**.

1. First you must analyse the data given to you in the table.
2. Identify which molecule contains single, double or triple bonds, and drag the appropriate label into the table.
3. Do the same for the number of electrons shared and bond type (single/double/triple).
4. From your analysis of the bond length and bond enthalpy values, identify if the arrow below the table shows the molecule in terms of increasing or decreasing bond length and bond enthalpy.
5. Drag the appropriate label into the table. Note that you do not need to use all four labels on the arrow.

Hydrocarbon	C_2H_6	C_2H_4	C_2H_2
Structural formula			
Bond between carbon atoms	single	double	triple
Bond length / pm	154	134	120
Bond enthalpy / KJ mol^{-1}	346	614	839
Number of electrons shared			
Bond order			

6 electrons

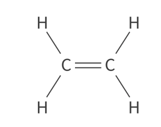
2 electrons

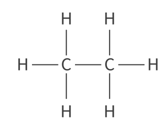
4 electrons

2

3

1





Check

6. **Interactive 4.** Comparison of single, double and triple bonds.

More information for interactive 4

Some of the gases you breathe in everyday demonstrate these different types of bonding. Nitrogen molecules, N_2, contain triple bonds. These are the strongest and shortest bonds.

Oxygen, O_2, and carbon dioxide, CO_2, molecules contain double bonds. These are of intermediate length and strength.

Hydrogen, H_2, and methane, CH_4, molecules contain single bonds – the longest and weakest of the three types of bonds.

This activity will allow you to model bond length and strength, using your fingers to represent the bonds. You should then apply this understanding in the second part, drawing Lewis formulas of different molecules and identifying the longest and strongest bonds in the molecules.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
  - Thinking skills — Reflecting on the credibility of results
  - Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20 minutes (10 minutes each part)
- **Activity type:** Pair activity (part 1); Individual activity (Part 2)

Part 1: Modelling bonds using your hands

In this activity, you will replicate single, double and triple bonds using the fingers on both hands.

1. To show single bonds, join the middle fingers on your left and right hands and push them together (**Figure 7**). Try to make the bond as long as possible. Ask a peer to 'break the bond' by gently slicing their hand through the joining.
2. To demonstrate double bonds, join the middle finger and the index fingers of each hand, again push them together. Notice what happens to the length of the bond when the index fingers join. Ask a peer to 'break the bond' by gently slicing their hand through the joining. Was this easier or more difficult now that two fingers were joined?
3. To demonstrate triple bonds, join the index and middle finger of each hand, and also the thumb, pushing together. Notice what has happened to the length of the bonds. Ask your peer to 'break the bond' again, and note whether it gets easier or more difficult as more fingers join.

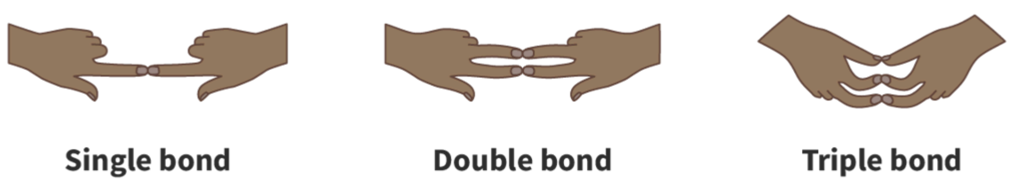


Figure 7. Demonstrate bonds using fingers of your left and right hand.

More information for figure 7

The image visually represents the concept of chemical bonds using hand gestures. There are three sections:

1. **Single bond**: Two hands are depicted coming together with the index fingers touching to represent a single bond. Beneath this, the text 'Single bond' is displayed.
2. **Double bond**: Here, the hands are shown with both the index and middle fingers touching, indicating a double bond between atoms. Below this visual, the text 'Double bond' is shown.
3. **Triple bond**: This is illustrated with the fingers forming a bridge using the index, middle, and ring fingers on both hands, symbolizing a triple bond. The text 'Triple bond' is featured underneath this depiction.

Each section illustrates a different type of bond, with the number of fingers touching reflecting the type of bond being represented.

[Generated by AI]

Part 2: Multiple bonds in Lewis formulas

You should complete this part individually and compare your answers with a peer.

Many of the planets which make up our solar system contain organic compounds, i.e. compounds composed of carbon and hydrogen. These include methane (CH_4), and small traces of ethane (C_2H_6), ethene (C_2H_4) and ethyne (HCCH).

Draw the Lewis formulas for each of these molecules.

Identify which bonds in each molecule will be longest and which ones will be strongest.